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July 19, 2026

Titanic - Machine Learning from Disaster

Data Workflow Project Summary

Overview

This is a data workflow project built around the Titanic - Machine Learning from Disaster dataset from Kaggle (<https://www.kaggle.com/c/titanic/data>). I loaded the dataset, cleaned up missing and unusable values, explored survival patterns across gender and passenger class, and built three visualizations to see what the data actually showed. The goal was to build a clean, repeatable workflow that could be used as a starting point for future machine learning work.

Dataset Description

The Titanic dataset contains information on 891 passengers from the 1912 sinking of the RMS Titanic. Each row represents one passenger and includes 15 columns covering things like age, sex, ticket class, fare paid, and whether they survived. The columns I focused on were survival, gender, passenger class, and age since those were the most interesting to look at in terms of who actually made it off the ship. The dataset had some quality issues, including 177 missing age values and a deck column that was missing for over 75% of passengers.

Workflow Description

Ingestion - I loaded the dataset using pandas `read_csv` and checked the first few rows to make sure everything came in correctly. I also ran a null check to see where the missing values were before doing anything else.

Cleaning - I wrote two functions to clean the data. The first one filled in the 177 missing age values using the median age. The second one dropped the deck column since most of it was empty and removed the two rows where the embarkation port was missing.

Exploratory Analysis - I wrote two functions to look at survival rates, one grouped by gender and one grouped by passenger class. The gender function to see if the women and children first line was real, and the passenger class function to see if even during an emergency, your wealth can have an impact on whether you live or die.

Visualizations - I made three charts. A bar chart for survival by gender, a bar chart for survival by passenger class, and a histogram showing age distribution split by whether passengers survived or not.

Summary - I wrote up what I found, what I thought was interesting, what the limitations were, and anything that surprised me. It was mostly what you'd expect from an incident that made saving the women and children first famous. I ran a Google search on the 80 year old survivor, and it turned out to be a known data error. The fact that there were posts and articles about it makes me feel like I was right to be suspicious of that outlier.

Key Decisions and Assumptions

I filled in the 177 missing age values with the median rather than dropping those rows since losing that much data would have hurt the analysis (Danchev, 2022). The deck column was dropped because over 75% of it was empty. For exploratory data analysis, I focused on gender and class because those were the most interesting questions in this dataset. Each chart was designed to show one thing: survival by gender, survival by class, and age distribution by survival.

Results and Interpretation

Chart 1 showed that female passengers survived at about 74% compared to 19% for males, which proves the women part of "women and children first" was followed during the evacuation. Chart 2 showed that first-class

passengers survived at 63% and second-class at 47%, while third-class was only 24%, suggesting that wealth played a role in who got off the ship. Chart 3 showed that the 15-30 age group took the most casualties by a wide margin, mostly because that was the largest group on the ship to begin with, and that children were also one of the few demographics that had a higher survival than casualty rate, which proved the other half of the “women and children first” line.

Responsible Practice (Bias and Data Quality)

The main bias risk in this workflow comes from filling in 177 missing age values with the median. Those filled-in values are not real and could skew any age-related analysis. It is also unclear where all the age data came from originally, since it could not have all been from ticket records. The Barkworth case proved that at least some of the ages in the dataset reflect when passengers died rather than how old they were on the ship, which is a known data quality issue (Purohit, 2021).

Reproducibility

Anyone who wants to rerun this project can clone the repository, install the dependencies using `pip install -r requirements.txt`, then open and run `data_workflow.ipynb` top to bottom. The `requirements.txt` file was generated using `pip freeze` to capture the exact package versions used. The project was built on a development branch and committed to GitHub in stages to show progress.

Sources and Citations

Danchev, V. (2022). Reproducible Data Science with Python: An Open Learning Resource. *Journal of Open Source Education*, 5(56), 156.

<https://doi.org/10.21105/jose.00156>

Purohit, K. (2021). Separation of Data Cleansing Concept from EDA.
International Journal of Data Science and Analysis, 7(3).

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